

LECTURE SHEET

EXERCISE-I

(Horizontal Motion)

LEVEL-I (MAIN)

Straight Objective Type Questions

- A boy walks to his school at a distance of 6km with constant speed of 2.5 kmph and walks back with a constant speed of 4 kmph. His average speed for round trip expressed in kmph is
1) 24/13 2) 40/13 3) 3 4) 1/2
- A particle moving with a constant acceleration describes in the last second of its motion $\frac{9}{25}$ of the whole distance. If it starts from rest, how long is the particle in motion and through what distance does it move if it describes 6 cm in the first sec.?
1) 5s ; 150 cm 2) 10 s ; 150 cm 3) 15 s ; 100 cm 4) None
- The reaction time for an automobile driver is 0.7 sec. If the automobile can be decelerated at 5 m/s^2 calculate the total distance travelled in coming to stop from an initial velocity of 8.33 m/s after a signal is observed.
1) 12.77 m 2) 14.82 m 3) 16.83 m 4) 19.65 m
- An express train moving at 30 m/s reduces its speed to 10 m/s in a distance of 240m. If the breaking force is increased by 12.5% in the beginning find the distance that it travels before coming to rest
1) 270 m 2) 240 m 3) 210 m 4) 195 m

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- The acceleration of a particle is increasing linearly with time t as bt . The particle starts from the origin with an initial velocity v_0 . The distance travelled by the particle in time t will be
a) $v_0t + \frac{1}{3}bt^2$ b) $v_0t + \frac{1}{2}bt^2$ c) $v_0t + \frac{1}{6}bt^3$ d) $v_0t + \frac{1}{3}bt^3$
- The speed of a train increases at a constant rate α from zero to v and then remains constant for an interval and finally decreases to zero at a constant rate β . If ℓ be the total distance described, and t the total time then
a) $t = \frac{\ell}{v} + \frac{v}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$ b) $t = \frac{\ell}{v} - \frac{v}{2} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)$ c) $t = \frac{\ell}{v} - \frac{v}{2} \left(\frac{1}{\alpha} - \frac{1}{\beta} \right)$ d) $t = \frac{\ell}{v} + \frac{v}{2} \left(\frac{1}{\alpha} - \frac{1}{\beta} \right)$
- The relation $3t = \sqrt{3x} + 6$ describes the displacement of a particle in one direction where x is in metres and t in sec. The displacement, when velocity is zero, is
a) 24 metres b) 12 metres c) 5 metres d) Zero

More than One correct answer Type Questions

- Starting from rest a particle is first accelerated for time t_1 with constant acceleration a_1 and then stops in time t_2 with constant retardation a_2 . Let v_1 be the average velocity in this case and s_1 the total displacement. In the second case, it is accelerated for the same time t_1 with constant acceleration $2a_1$ and comes to rest with constant retardation a_2 in time t_3 . If v_2 is the average velocity in this case and s_2 the total displacement. Then
a) $v_2 = 2v_1$ b) $2v_1 < v_2 < 4v_1$ c) $s_2 = 2s_1$ d) $2s_1 < s_2 < 4s_1$

5. A particle starts from rest and travels with constant acceleration 'a'. After a time of ' t_0 ' its acceleration direction gets reversed. Identify correct options.
- It's velocity when it returns back to initial point is $\sqrt{2}at_0$
 - It's velocity when it returns back to initial point is $2at_0$
 - Total distance travelled by particle when it reaches back to initial point is $2at_0^2$
 - Time taken to come back to initial point $(2+\sqrt{2})t_0$

Integer Type Questions

6. A car starts from rest with an acceleration 6m/s^2 which decreases to zero linearly with time, in 10 second, after which the car continues to travel with constant speed. The time required for the car to travel 400m from the start is $16.67t$ sec. The value of ' t ' is
7. A train stops at two stations s distance apart and takes time t on the journey from one station to the other. Its motion is first of uniform acceleration a and then immediately of uniform retardation b , then the relation between a , b , s , t is $\frac{1}{a} + \frac{1}{b} = \frac{kt^2}{2s}$. The value of ' k ' is
8. Velocity-time equation of a particle moving in a straight line is $v=2t-4$ for $t \leq 2\text{s}$ and $v=4-2t$ for $t > 2\text{s}$. The distance travelled (in meters) by the particle in the time interval from $t=0$ to $t=4\text{s}$ is $4k$. The value of ' k ' is (Here t is in second and v in m/s) :

EXERCISE-II**(Vertical Motion)****LEVEL-I (MAIN)****Straight Objective Type Questions**

- A freely falling body travels ___ of total distance in 5th second (which is last second)
 - 8%
 - 12%
 - 25%
 - 36%
- A splash is heard 3.12s after a stone is dropped into a well 45m deep. The speed of sound in air is [$g=10\text{ms}^{-2}$]
 - 330 ms^{-1}
 - 375 ms^{-1}
 - 340 ms^{-1}
 - 346 ms^{-1}
- A ball dropped on to the floor from a height of 10m rebounds to a height of 2.5m. If the ball is in contact with the floor for 0.02s, its average acceleration during contact is
 - 2100 ms^{-2}
 - 1050 ms^{-2}
 - 4200 ms^{-2}
 - 9.8 ms^{-2}
- A vertical disc has three grooves directed along chords AB, AC and AD. Three bodies begin to slide down the respective grooves from A simultaneously. If $AB > AC > AD$, the respective time intervals to reach the bottoms of the respective grooves t_1, t_2 and t_3 are
 - $t_1 > t_2 > t_3$
 - $t_1 < t_2 < t_3$
 - $t_1 > t_2 < t_3$
 - $t_1 = t_2 = t_3$
- If the distance travelled by a freely falling body in the last second of its journey is equal to the distance travelled in the first 2s, the time of descent of the body is
 - 5 s
 - 1.5 s
 - 2.5 s
 - 3 s
- The distance travelled by a body during last second of its total flight is d when the body is projected vertically up with certain velocity. If the velocity of projection is doubled, the distance travelled by the body during last second of its total flight is
 - $2d$
 - d
 - $2d + \frac{g}{2}$
 - $2d - \frac{g}{2}$

7. If a stone dropped from the top of a tower travels half of the height of the tower during last second of its fall, the time of fall is (in seconds)
 1) $3 + \sqrt{2}$ 2) $4 + \sqrt{2}$ 3) $2 - \sqrt{2}$ 4) $2 + \sqrt{2}$
8. A helicopter is ascending vertically with a speed of 8.0 ms^{-1} . At a height of 120 m above the earth, a package is dropped from a window. How much time does it take for the package to reach the ground ?
 1) 1.23s 2) 3.23s 3) 5.83s 4) 7.83s
9. A ball dropped from a point P crosses a point Q in t seconds. The time taken by it to travel from Q to R, if $PQ = QR$
 1) t 2) $\sqrt{2} t$ 3) $2 t$ 4) $(\sqrt{2} - 1) t$
10. A girl is standing on the top edge of an 18 m high building. She tosses a coin upward with a speed of 7.0 ms^{-1} . How long does it take for the coin to hit the ground ? How fast is the coin going just before it strikes the ground ? ($g = 10 \text{ ms}^{-2}$)
 1) 2.70s, 20 m/s 2) 1.70s, 30 m/s 3) 4.70s, 50m/s 4) 10s, 50 m/s

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. From an elevated point P a stone is projected vertically upward. When it reaches a distance y below the point of projection its velocity is double the velocity when it was at a height y above p. The greatest height reached by it above P is
 a) $\frac{2y}{3}$ b) $\frac{5y}{3}$ c) $\frac{y}{3}$ d) $2y$
2. Ball A is dropped from the top of a building. At the same instant another identical ball B is thrown vertically upward from the ground. When the balls collide, they are moving in opposite directions and the speed of A is twice the speed of B. At what fraction of the height of the building did the collision occur
 a) $2/3$ b) $4/3$ c) $5/3$ d) $1/3$
3. A body P is thrown vertically up with velocity 30 ms^{-1} and another body Q is thrown up along the same vertically line with the same velocity but 1 second later from the ground. When they meet ($g = 10 \text{ ms}^{-2}$)
 a) P travels for 2.5 s b) Q travels for 3.5s c) P travels for 3.5 s d) Q travels for 1s
4. A parachutist drops freely from an aeroplane for 8s before the parachute opens out. Then he descends with a net retardation of 2 ms^{-2} reaching the ground with a velocity of 6 ms^{-1} . The height from which he bails out of the aeroplane is ($g = 10 \text{ ms}^{-2}$)
 a) 1929 m b) 1911 m c) 2195 m d) 2000m

Linked Comprehension Type Questions

Passage-I :

A balloon rises from rest on the ground with acceleration $g/8$. A stone is dropped when balloon is at a height of 39.2m

5. The taken by stone to reach ground is
 a) 2 sec b) 4 sec c) 6 sec d) 3 sec
6. Velocity of stone when it strikes ground is
 a) $2g \text{ m/s}$ b) $4g \text{ m/s}$ c) $3g \text{ m/s}$ d) $5g \text{ m/s}$

Matrix Matching Type Questions

7. A balloon rises up (initial velocity is zero) with constant net acceleration of 10 m/s^2 . After 2 seconds a body drops from the balloon. After further 2 seconds match the Column-I with Column-II. ($g = 10 \text{ m/s}^2$)

Column-I

- A) Height of body from the ground
B) Speed of the body
C) Displacement of the body in 2s after it drops from the balloon
D) Acceleration of the body

Column-II

- P) Zero
Q) 10 SI units
R) 40 SI units
S) 20 SI units

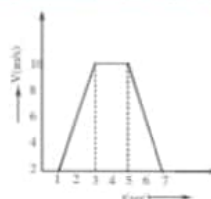
Integer Type Questions

8. A particle is projected vertically upwards with speed 'u' in a medium in which the resistance to motion is proportional to square of speed. Initially net force is double to its weight. The time of ascent and maximum height are $\frac{k\pi u}{4g}$, $\frac{pu^2}{4g} \log(2)$. The values of 'k and p' are

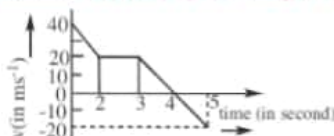
EXERCISE-III**(Graphs)****LEVEL-I (MAIN)**Straight Objective Type Questions

1. For the velocity-time graph shown in figure below the distance covered by the body in last two seconds of its motion is what fraction of the total distance covered by it in all the total distance covered by it in all the seven seconds

- 1) $\frac{1}{2}$ 2) $\frac{1}{4}$
3) $\frac{1}{3}$ 4) $\frac{2}{3}$

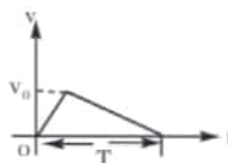


2. In the given v - t graph, the distance travelled by the body in 5 second will be

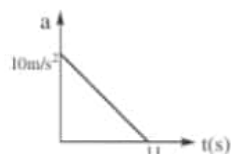


- 1) 20m 2) 40m 3) 80m 4) 100m
3. A particle moves in a straight line obeying the v-t graph as shown in the figure. The average velocity of the particle over the time T is :

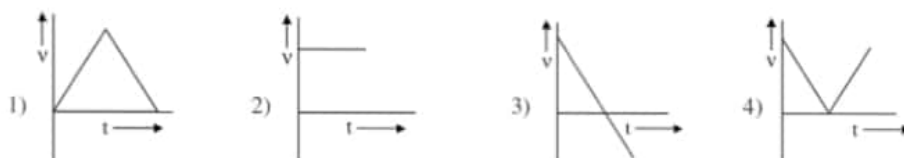
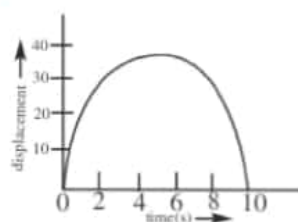
- 1) zero 2) $\frac{v_0}{2}$
3) $\frac{2v_0}{3}$ 4) $\frac{3v_0}{4}$



4. A particle starts from rest. Its acceleration (a) versus time (t) is as shown in the figure. The maximum speed of the particle will be



- 1) 110 m/s 2) 55 m/s 3) 550 m/s 4) 660 m/s
5. The displacement-time graph of a moving object is shown in figure. Which of the velocity-time graphs shown in figure could represent the motion of the same body ?

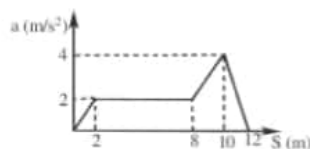


6. In the x - t plot of a particle in one dimensional motion, four different equal intervals of time are shown. In which interval, average speed is greatest



- 1) 1 2) 2 3) 3 4) 4
7. In above problem identify incorrect sign of average velocity for an interval
- 1) 1 $\rightarrow$ + 2) 2 $\rightarrow$ - 3) 3 $\rightarrow$ - 4) 4 $\rightarrow$ +
8. The acceleration displacement graph of a particle moving in a straight line is shown in the figure. Initial velocity of the particle is zero. Find the velocity of the particle when displacement of the particle is $S = 12$ m

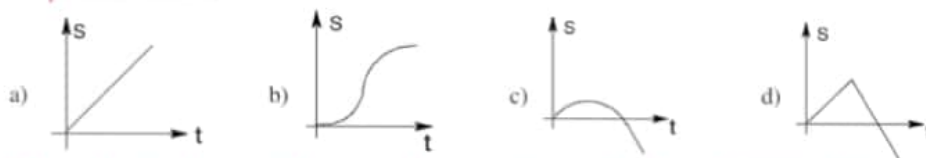
- 1) $3\sqrt{2}$ m/s
2) $2\sqrt{5}$ m/s
3) 4 m/s
4) $4\sqrt{3}$ m/s



LEVEL-II (ADVANCED)

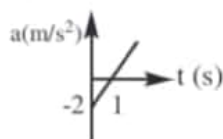
Straight Objective Type Questions

1. A particle is moving in x-y plane with $y = \frac{x}{2}$ and $v_x = 4 - 2t$. The displacement versus time graph of the particle would be

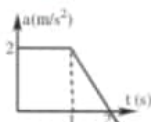


2. The acceleration of a particle varies with time as shown in the figure. Find the displacement of the particle in the interval from $t = 2\text{ s}$ to $t = 4\text{ s}$. Assume that $v = 0$ at $t = 0$

- a) 6.67 m
b) 5.67 m
c) 4.67 m
d) 3.67 m



3. The acceleration - time graph of a particle moving along a straight line (starting from rest) is shown. After what time the particle attains its initial position?

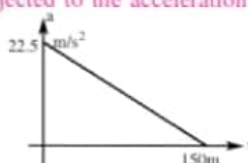


- a) 2.8 sec b) 4.28 sec c) 5 sec d) 3 sec

More than One correct answer Type Questions

4. A jet plane starts from rest at $S = 0$ and it is subjected to the acceleration shown

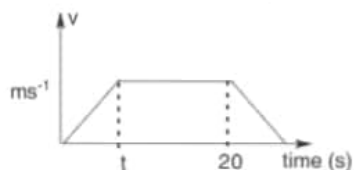
- a) Velocity at $S = 60\text{ m}$ is 46.47 m/s
b) Velocity at $S = 150\text{ m}$ is 58.1 m/s
c) Acceleration at $S = 60$ is 13.5 m/s^2
d) Velocity will be maximum at $S = 75$



Linked Comprehension Type Questions

Passage-I :

The velocity-time graph of a particle moving along a straight line is as shown in fig. The rate of acceleration and deceleration is constant and it is equal to 5 ms^{-2} . If the average velocity during the motion is 20 ms^{-1} , then



5. The value of t is

- a) 5 s b) 10 s c) 20 s d) $5\sqrt{2}\text{ s}$

6. The maximum velocity of the particle is
 a) 20 m/s b) 25 m/s c) 30 m/s d) 40 m/s
7. The distance travelled with uniform velocity is
 a) 375m b) 125 m c) 300 m d) 450 m

Matrix Matching Type Questions

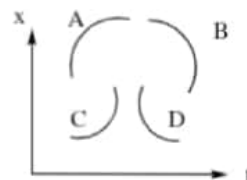
8. Figure shows the position –time graph of particles moving along a straight line. Match the entries of Column I with the entries of Column II

Column-I

- A) The particle A is
 B) The particle B is
 C) The particle C is
 D) The particle D is

Column-II

- P) Accelerating
 Q) Decelerating
 R) Speeding up
 S) Slowing down

**EXERCISE-IV***(2D other than Projectile)***LEVEL-I (MAIN)****Straight Objective Type Questions**

1. A particle moving with a velocity equal to 0.4 ms^{-1} is subjected to an acceleration of 0.15 ms^{-2} for 2 seconds in a direction at right angles to the direction of motion. The magnitude of the final velocity is
 1) 0.3 ms^{-1} 2) 0.4 ms^{-1} 3) 0.5 ms^{-1} 4) 0.7 ms^{-1}
2. Two boys are standing at the ends A and B of a ground where $AB = a$. The boy at B starts running in a direction perpendicular to AB with velocity v_1 . The boy at A starts running simultaneously with velocity v and catches the other boy in a time t , where t is
 1) $a/\sqrt{v^2 + v_1^2}$ 2) $\frac{a}{\sqrt{v^2 - v_1^2}}$ 3) $a/(v - v_1)$ 4) $a/(v + v_1)$
3. Velocity of a particle at time $t = 0$ is 2 ms^{-1} . A constant acceleration of 2 ms^{-2} acts on the particle for 1 second at an angle of 60° with its initial velocity. Find the magnitude of velocity at the end of 1 second.
 1) $\sqrt{3} \text{ m/s}$ 2) $2\sqrt{3} \text{ m/s}$ 3) 4 m/s 4) 8 m/s

LEVEL-II (ADVANCED)**Straight Objective Type Questions**

1. A point moves in xy-plane according to equation $x=at$, $y = at(1-bt)$ where a and b are positive constants and t is time. The instant at which velocity vector is at $\pi/4$ with acceleration vector is given by:
 a) $1/a$ b) $1/b$ c) $1/a + 1/b$ d) $(a+b)/(a^2+b^2)$
2. A particle moves in x-y plane according to the law $x = 4 \sin 6t$ and $y = 4(1-\cos 6t)$. The distance traversed by the particle in 4 seconds is: (x and y are in meters)
 a) 96 m b) 48 m c) 24 m d) 108 m

Integer Type Questions

3. A particle move along the parabolic path $x = y^2 + 2y + 2$ in such a way that the y-component of velocity vector remain 5m/s during the motion. The magnitude of the acceleration of the particle is $k(25)\text{m/sec}$. where 'k' is
4. A ball is projected inside a room of height 7m with a speed of 20m/s and at an angle of 37° with horizontal. If its collision with ceiling results in reversal of direction only of y-component of velocity, what is the total time of flight (in sec) till it strikes floor.....

EXERCISE-V*(Oblique Projectile & Projecting on incline plane)***LEVEL-I (MAIN)**Straight Objective Type Questions

- The speed of a projectile at its maximum height is $\sqrt{3}/2$ times its initial speed. If the range of the projectile is p times the maximum height attained by it, then p =
 1) $4/3$ 2) $2\sqrt{3}$ 3) $4\sqrt{3}$ 4) $3/4$
- Two bodies are thrown with the same initial velocity at angles α and $(90-\alpha)$ to the horizon. What is the ratio of the maximum heights reached by the bodies.
 1) $\cot^2 \alpha$ 2) $\tan^2 \alpha$ 3) $\sec^2 \alpha$ 4) $\cos^2 \alpha$
- Two balls are projected simultaneously in same verticle plane from the same point with velocities V_1 and V_2 with angles θ_1 and θ_2 respectively with the horizontal. If $V_1 \cos \theta_1 = V_2 \cos \theta_2$, the path of one ball as seen from the position of another ball is
 1) parabola 2) horizontal straight line
 3) vertical straight line 3 4) straight line making 45° with vertical
- The height y and horizontal distance x covered by a projectile in a time t seconds are given by the equations $y = 8t - 5t^2$ and $x = 6t$. If x and y are measured in metres, the velocity of projection is
 1) 8 ms^{-1} 2) 6 ms^{-1} 3) 14 ms^{-1} 4) 10 ms^{-1}
- A body is projected with a velocity u at an angle of 60° to the horizontal. The time interval after which it will be moving in a direction of 30° to the horizontal is
 1) $\frac{u}{\sqrt{3}g}$ 2) $\frac{\sqrt{3}u}{g}$ 3) $\frac{\sqrt{3}u}{2g}$ 4) $\frac{2u}{\sqrt{3}g}$
- A player kicks a foot ball obliquely at a speed of 20ms^{-1} so that its range is maximum. Another player at a distance of 24m away in the direction of kick starts running at that instant to catch the ball. Before the ball hits the ground to catch it, the speed with which the second player has to run is ($g = 10\text{ms}^{-2}$)
 1) 4 ms^{-1} 2) $4\sqrt{2} \text{ ms}^{-1}$ 3) $8\sqrt{2} \text{ ms}^{-1}$ 4) 8 ms^{-1}
- A projectile has initially the same horizontal velocity as it would acquire if it had moved from rest with uniform acceleration of 3ms^{-2} for 0.5 minutes. If the maximum height reached by it is 80m then the angle of projection is [$g = 10\text{ms}^{-2}$]
 1) $\tan^{-1}(3)$ 2) $\tan^{-1}\left(\frac{3}{2}\right)$ 3) $\tan^{-1}\left(\frac{4}{9}\right)$ 4) $\sin^{-1}\left(\frac{4}{9}\right)$

Integer Type Questions

3. A particle move along the parabolic path $x = y^2 + 2y + 2$ in such a way that the y-component of velocity vector remain 5m/s during the motion. The magnitude of the acceleration of the particle is $k(25)\text{m/sec}$. where 'k' is
4. A ball is projected inside a room of height 7m with a speed of 20m/s and at an angle of 37° with horizontal. If its collision with ceiling results in reversal of direction only of y-component of velocity, what is the total time of flight (in sec) till it strikes floor.....

EXERCISE-V*(Oblique Projectile & Projecting on incline plane)***LEVEL-I (MAIN)**Straight Objective Type Questions

1. The speed of a projectile at its maximum height is $\sqrt{3}/2$ times its initial speed. If the range of the projectile is p times the maximum height attained by it, then p =
 1) $4/3$ 2) $2\sqrt{3}$ 3) $4\sqrt{3}$ 4) $3/4$
2. Two bodies are thrown with the same initial velocity at angles α and $(90-\alpha)$ to the horizon. What is the ratio of the maximum heights reached by the bodies.
 1) $\cot^2 \alpha$ 2) $\tan^2 \alpha$ 3) $\sec^2 \alpha$ 4) $\cos^2 \alpha$
3. Two balls are projected simultaneously in same verticle plane from the same point with velocities V_1 and V_2 with angles θ_1 and θ_2 respectively with the horizontal. If $V_1 \cos \theta_1 = V_2 \cos \theta_2$, the path of one ball as seen from the position of another ball is
 1) parabola 2) horizontal straight line
 3) vertical straight line 3 4) straight line making 45° with vertical
4. The height y and horizontal distance x covered by a projectile in a time t seconds are given by the equations $y = 8t - 5t^2$ and $x = 6t$. If x and y are measured in metres, the velocity of projection is
 1) 8 ms^{-1} 2) 6 ms^{-1} 3) 14 ms^{-1} 4) 10 ms^{-1}
5. A body is projected with a velocity u at an angle of 60° to the horizontal. The time interval after which it will be moving in a direction of 30° to the horizontal is
 1) $\frac{u}{\sqrt{3}g}$ 2) $\frac{\sqrt{3}u}{g}$ 3) $\frac{\sqrt{3}u}{2g}$ 4) $\frac{2u}{\sqrt{3}g}$
6. A player kicks a foot ball obliquely at a speed of 20ms^{-1} so that its range is maximum. Another player at a distance of 24m away in the direction of kick starts running at that instant to catch the ball. Before the ball hits the ground to catch it, the speed with which the second player has to run is ($g = 10\text{ms}^{-2}$)
 1) 4 ms^{-1} 2) $4\sqrt{2} \text{ ms}^{-1}$ 3) $8\sqrt{2} \text{ ms}^{-1}$ 4) 8 ms^{-1}
7. A projectile has initially the same horizontal velocity as it would acquire if it had moved from rest with uniform acceleration of 3ms^{-2} for 0.5 minutes. If the maximum height reached by it is 80m then the angle of projection is [$g = 10\text{ms}^{-2}$]
 1) $\tan^{-1}(3)$ 2) $\tan^{-1}\left(\frac{3}{2}\right)$ 3) $\tan^{-1}\left(\frac{4}{9}\right)$ 4) $\sin^{-1}\left(\frac{4}{9}\right)$

8. A particle is projected with velocity $2\sqrt{gh}$ so that it just clears two walls of equal height h which are a distance $2h$ from each other. The time interval for which the particle travels between this two walls is ($\theta = 60^\circ$)
- 1) $2\sqrt{\frac{h}{g}}$ 2) $\sqrt{\frac{h}{g}}$ 3) $\sqrt{\frac{2h}{g}}$ 4) $\sqrt{\frac{h}{2g}}$
9. A particle is projected with a certain velocity at an angle α above the horizontal from the foot of an inclined plane of inclination 30° . If the particle strikes the plane normally then α is equal to
- 1) $30^\circ + \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ 2) 45° 3) 60° 4) $30^\circ + \tan^{-1} 2\sqrt{3}$

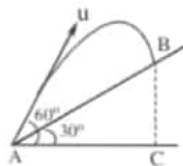
LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. A ball is thrown with velocity u making an angle θ with the horizontal. Its velocity vector is normal to initial velocity (u) vector after a time interval of
- a) $\frac{u \sin \theta}{g}$ b) $\frac{u}{g \cos \theta}$ c) $\frac{u}{g \sin \theta}$ d) $\frac{u \cos \theta}{g}$
2. A body is projected with velocity u making an angle θ with the horizontal. Its velocity when it is perpendicular to the initial velocity vector is
- a) $u \sin \theta$ b) $u \cot \theta$ c) $u \tan \theta$ d) $u \cos \theta$
3. A particle is projected from the ground with velocity u making an angle θ with the horizontal. At half of its maximum heights,
- a) its horizontal velocity is $u \cos \theta$ b) its vertical velocity is $\frac{u \sin \theta}{\sqrt{2}}$
c) its velocity is $u \left(\frac{1 + \cos^2 \theta}{2} \right)^{\frac{1}{2}}$ d) all the above are true
4. A particle is aimed at a mark which is in the same horizontal plane as that of point of projection. It falls 10 m short of the target when it is projected of an angle of 75° and falls 10m ahead of the target when it is projected with an elevation of 45° . The angle of projection for which the particle exactly hits this target is ($g=10\text{ms}^{-1}$)
- a) $\frac{1}{2} \sin^{-1}\left(\frac{3}{4}\right)$ b) $\frac{1}{2} \sin^{-1}\left(\frac{4}{5}\right)$ c) $\frac{1}{2} \tan^{-1}\left(\frac{1}{2}\right)$ d) $\tan^{-1}(2)$
5. From a point on the ground at a distance 2m from the foot of a vertical wall, a ball is thrown at an angle of 45° of which just clears the top, of the wall and afterward strikes the ground at a distance 4m on the otherside. The height of the wall is
- a) $2/3$ m b) $3/4$ m c) $1/2$ m d) $4/3$ m
6. When a body is projected from a level ground the ratio of its speed in the vertical and horizontal direction is 4 : 3. If the velocity of projection is u , the time after which, the ratio of the velocities in the vertical and horizontal directions are reversed is
- a) $\frac{7u}{20g}$ b) $\frac{35u}{10g}$ c) $\frac{9u}{g}$ d) $\frac{10u}{g}$

7. In the figure, the time taken by the projectile to reach from A to B is t . Then the distance AB is equal to

- a) $\frac{ut}{\sqrt{3}}$ b) $\frac{\sqrt{3}ut}{2}$
c) $\sqrt{3}ut$ d) $2ut$



8. A ball is projected horizontal with a speed v from the top of a plane inclined at an angle 45° with the horizontal. How far from the point of projection with the ball strike the plane?

- a) $\frac{v^2}{g}$ b) $\sqrt{2}\frac{v^2}{g}$ c) $\frac{2v^2}{g}$ d) $\sqrt{2}\left[\frac{2v^2}{g}\right]$

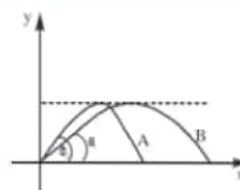
9. A plane surface is inclined making an angle θ with the horizontal. From the bottom of this inclined plane, a bullet is fired with velocity ' v '. Find the maximum possible range of the bullet on the inclined plane

- a) $\frac{v^2}{g}$ b) $\frac{v^2}{g(1+\cos\theta)}$ c) $\frac{v^2}{g(1+\sin\theta)}$ d) $\frac{v^2}{g(1-\sin\theta)}$

More than One correct answer Type Questions

10. Two particles are thrown from the same point in the same vertical plane, as shown in figure simultaneously so as to reach the same maximum height. Then indicate the correct statements :

- a) Time of flight for B is less than that of A
b) Projection speed of B is greater than that of A
c) Horizontal component of velocity for B is greater than that of A
d) The vertical component of velocities of both A and B are always equal throughout the duration for which both the particles are in air.

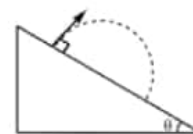


11. The horizontal distance x and the vertical height y of a projectile at time t are given by $x = at$ and $y = bt^2 + ct$ where a , b and c are constants. Then

- a) The speed of the projectile 1 second after it is fired is $(a^2 + b^2 + c^2)^{1/2}$
b) The angle with the horizontal at which the projectile is fired is $\tan^{-1}(c/a)$
c) The acceleration due to gravity is $-2b$
d) The initial speed of the projectile is $(a^2 + c^2)^{1/2}$

12. A projectile is fired with a velocity ' u ' at right angles to an inclined plane which is at an angle ' θ ' with horizontal

- a) Range on inclined plane is $\frac{2u^2}{g} \tan\theta \sec\theta$
b) Time of flight is $\frac{2u}{g \cos\theta}$
c) The motion of particle along any direction is uniformly accelerated motion
d) Horizontal motion is uniform motion



13. Particle A is released from a point P on a smooth inclined plane inclined at an angle α with the horizontal. At the same instant another particle B is projected with initial velocity U making an angle β with the horizontal. Both the particles meet again on the inclined plane.

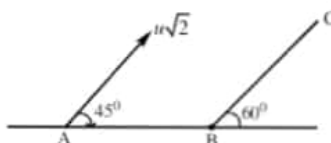
- Velocity along the inclined plane is same for both particles always
- Acceleration along the inclined plane is same for both
- $\alpha + \beta = 90^\circ$
- Relative velocity between the particles first decreases and later increases



Linked Comprehension Type Questions

Passage-I :

A particle projected from a point A with velocity $u\sqrt{2}$ at an angle 45° with horizontal. It strikes the plane BC at right angle



14. The velocity with which it strikes the plane is

- $\frac{u}{\sqrt{3}}$
- $\frac{2u}{\sqrt{3}}$
- $\frac{\sqrt{3}u}{2}$
- u

15. Time taken for the collision is

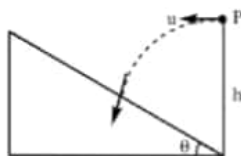
- $\frac{(\sqrt{3}+1)u}{g}$
- $\frac{(\sqrt{3}+1)u}{\sqrt{3}g}$
- $\frac{(\sqrt{3}+1)u}{2g}$
- u

16. Change in velocity is

- u
- $\frac{\sqrt{3}u}{2}$
- $\frac{(\sqrt{3}+1)u}{2\sqrt{3}}$
- $\frac{(\sqrt{3}+1)u}{\sqrt{3}}$

Passage-II :

A particle projected horizontally from P, so that it hits the inclined plane perpendicularly



17. $U =$

- $\sqrt{\frac{2gh \sin^2 \theta}{1 + \cos^2 \theta}}$
- $\sqrt{\frac{2gh \cos^2 \theta}{1 + \cos^2 \theta}}$
- $\sqrt{\frac{2gh \sin^2 \theta}{1 + \sin^2 \theta}}$
- $\sqrt{\frac{2gh \cos^2 \theta}{1 + \sin^2 \theta}}$

18. Time of journey is

- $\sqrt{\frac{2h \cos^2 \theta}{g(1 + \sin^2 \theta)}}$
- $\sqrt{\frac{2h \sin^2 \theta}{g(1 + \sin^2 \theta)}}$
- $\sqrt{\frac{2h \cos^2 \theta}{g(1 + \cos^2 \theta)}}$
- $\sqrt{\frac{2h}{g \cos \theta}}$

Matrix Matching Type Questions

19. The magnitude of velocity of projectile and time are related as $\frac{v^2}{a^2} - \left(t - \frac{b}{a}\right)^2 = \frac{c^2}{a^2}$ then

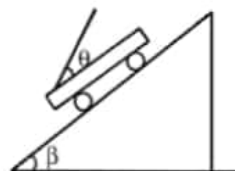
Column-I**Column-II**

- | | |
|-------------------------------|-------------|
| A) Range | P) C |
| B) Height | Q) $2b/a$ |
| C) Time of flight | R) $2bc/a$ |
| D) Velocity at maximum height | S) $b^2/2a$ |
| | T) ab/c |

20. A cannon is fixed with a smooth massive trolley car at an angle ' θ ' as shown in diagram. The trolley car slides from rest down the inclined plane of inclination ' β '. Muzzle velocity of shell fired from cannon is ' U '. t_1, t_2 are time instants at which shell-1, shell-2 are fired such that they move perpendicular to inclined plane and vertically upwards respectively.

Column-I**Column-II**

- | | |
|------------------------------|--|
| A) $t_1 =$ | P) $\frac{2u}{g}$ |
| B) $t_2 =$ | Q) $\frac{2u \sin \theta}{g \cos \beta}$ |
| C) time of flight of shell-1 | R) $\frac{u \cos \theta}{g \sin \beta}$ |
| D) time of flight of shell-2 | S) $\frac{2u \cos(\beta + \theta)}{g \sin 2\beta}$ |
| | T) $\frac{u}{g \sin \beta}$ |

Integer Type Questions

21. A ground to ground projectile projected at $t = 0$ is at points A at $t = T/3$, is at point B at $t = 5T/6$ and reaches the ground at $t = T$. The difference in heights between points A and B is $gT^2/8k$. The value of ' k ' is _____

EXERCISE-VI*(Horizontal Projectile)***LEVEL-I (MAIN)**Straight Objective Type Questions

1. A body is projected horizontally from a height of 78.4 m with a velocity 10 ms^{-1} . Its velocity after 3 seconds is : [$g = 10 \text{ ms}^{-2}$] (Take direction of projection on $\hat{i}$ and vertically upward direction on $\hat{j}$).
- 1) $10\hat{i} - 30\hat{j}$ 2) $10\hat{i} + 30\hat{j}$ 3) $20\hat{i} - 30\hat{j}$ 4) $10\hat{i} + 10\sqrt{3}\hat{j}$

2. A ball is projected horizontally from the top of a building 19.6 m high. If the line joining the point of projection to the point where it hits the ground makes an angle of 45° to the horizontal, the initial velocity of the ball is
 1) 4.9 ms^{-1} 2) 9.8 ms^{-1} 3) 19.6 ms^{-1} 4) 14.7 ms^{-1}
3. Two thin wood screens A and B are separated by 200m. A bullet travelling horizontally at a speed of 600 ms^{-1} hits the screen A, penetrates through it and finally emerges out from B making holes in A and B. If the resistance of air and wood are negligible, the difference of heights of the holes in A and B is
 1) 5 m 2) $\frac{49}{90} \text{ m}$ 3) $\frac{7}{\sqrt{90}} \text{ m}$ 4) zero
4. The height and width of each step of a staircase are 20cm and 30cm respectively. A ball rolls off the top of a stair with horizontal velocity v and hits the fifth step. The magnitude of v is ($g=10 \text{ ms}^{-2}$)
 1) $1.5\sqrt{5} \text{ ms}^{-1}$ 2) $3\sqrt{5} \text{ ms}^{-1}$ 3) 7.5 ms^{-1} 4) 1.5 ms^{-1}
5. A ball thrown horizontally with velocity v from the top of a tower of height h reaches the ground in t seconds. If another ball of double the mass is thrown horizontally with velocity $3v$ from the top of another tower of height $4h$ it reaches the ground in (seconds)
 1) $3t/4$ 2) $t/2$ 3) $4t$ 4) $2t$

LEVEL-II (ADVANCED)

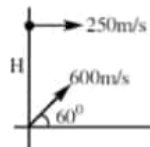
Straight Objective Type Questions

1. A fighter plane flying horizontally at an altitude of 2 km with speed of 540 kmph passes directly over head an antiaircraft gun. If the gun can fire a bullet at the muzzle speed of 500 ms^{-1} , at what angle with the vertical the gun should fire the bullet so that the bullet hits the plane ?
 a) $\cos^{-1}(3/10)$ b) $\sin^{-1}(3/10)$ c) $\tan^{-1}(10/3)$ d) 45°
2. At a certain height a body at rest explodes into two equal fragments with one fragment receiving a horizontal velocity of 10 ms^{-1} . The time interval after the explosion for which the velocity vectors of the two fragments become perpendicular to each other is ($g=10 \text{ ms}^{-2}$)
 a) 1s b) 2 s c) 1.5 s d) 1.75 s
3. In this above problem the horizontal distance between the two fragments when their velocity vectors are perpendicular to each other is
 a) 10 m b) 20m c) 40m d) 50m
4. In the above problem, the time taken by the displacement vectors of the two fragments to become perpendicular to each other is
 a) 1s b) 1.5 s c) 2s d) 4s
5. At a certain height a shell at rest explodes into two equal fragments. One of the fragments receives a horizontal velocity u . The time interval after which, the velocity vectors will be inclined at 120° to each other is
 a) $\frac{u}{\sqrt{3}g}$ b) $\frac{\sqrt{3}u}{g}$ c) $\frac{2u}{\sqrt{3}g}$ d) $\frac{u}{2\sqrt{3}g}$

6. A bomb at rest at the summit of a cliff breaks into two equal fragments. One of the fragments attains a horizontal velocity of $20\sqrt{3} \text{ ms}^{-1}$. The horizontal distance between the two fragments, when their displacement vectors is inclined at 60° relative to each other is ($g = 10 \text{ ms}^{-2}$)
- a) $40\sqrt{3} \text{ m}$ b) $80\sqrt{3} \text{ m}$ c) $120\sqrt{3} \text{ m}$ d) $480\sqrt{3} \text{ m}$
7. Two particles are projected horizontally from the same elevated point in opposite directions with velocities 4 ms^{-1} and 9 ms^{-1} respectively. At the moment when their velocity vectors are mutually perpendicular, the separation between them is ($g = 10 \text{ ms}^{-2}$)
- a) 5 m at angle $\tan^{-1} \frac{9}{4}$ b) 3 m at angle $\tan^{-1} \frac{9}{4}$
 c) 7.8 m horizontally d) 5 m horizontally

Integer Type Questions

8. A fighter plane enters inside enemy territory, at time $t=0$, with velocity 250 m/s and moves horizontally with constant acceleration $a = 20 \text{ m/s}^2$. An enemy tank at border, spot the plane and fires shots at an angle 60° with horizontal and with velocity $U = 600 \text{ m/s}$. The altitude H of plane it can be hit by the shot is $2473h \text{ m}$. The value of ' h ' is



EXERCISE-VII

(Relative motion)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. A person in lift which ascends up with acceleration 10 ms^{-2} drops a stone from a height 10 m . The time of decent is __ [$g=10 \text{ ms}^{-2}$]
- 1) 1 s 2) 2 s 3) 15 m 4) 20 m
2. Find the speed of two objects if, when they move uniformly towards each other they get 4 m closer in every sec and when they move uniformly in the same direction with their original speeds, they get 4 m closer every 10 s
- 1) $1.8 \text{ m/s}, 1.6 \text{ m/s}$ 2) $2 \text{ m/s}, 1.8 \text{ m/s}$ 3) $2.2 \text{ m/s}, 1.8 \text{ m/s}$ 4) $1.6 \text{ m/s}, 1.4 \text{ m/s}$
3. Two particles start moving from the same point along the same straight line. The first moves with constant velocity v and the second with constant acceleration a . During the time that elapses before the second catches the first, the greatest distance between the particles is
- 1) $\frac{V^2}{a}$ 2) $\frac{V^2}{2a}$ 3) $\frac{2V^2}{a}$ 4) $\frac{V^2}{4a}$
4. Two cars are moving in the same direction with a speed of 30 km h^{-1} . They are separated from each other by 5 km . Third car moving in the opposite direction meets the two cars after an interval of 4 minutes . What is the speed of the third car?
- 1) 30 kmh^{-1} 2) 35 kmh^{-1} 3) 40 kmh^{-1} 4) 45 kmh^{-1}

5. Two ships are 10 km apart on a line running south to north. The one farther north is streaming west at 40 km per hr, the other ship is streaming towards north with the same speed. The time after which the two ships are at closest distance is
 1) 7.5 min 2) 12 min 3) 14 min 4) 20 min
6. Two ships are 10 km apart on a line from South to North. The one further North is moving towards West at 40 km/hr and other is moving towards North at 40 km/hr. The distance of closest approach is
 1) 5 km 2) $5\sqrt{2}$ km 3) 10 km 4) $\frac{5}{\sqrt{2}}$ km
7. The distance between two moving particles at any time is a . If v be their relative velocity and v_1 and v_2 be the components of v along and perpendicular to a . The time when they are closest to each other is
 1) $\frac{av_1}{v^2}$ 2) $\frac{av_2}{v^2}$ 3) $\frac{av}{v_1^2}$ 4) $\frac{av}{v_2^2}$
8. A boat crosses a river of width 200m in the shortest time and is found to experience a drift of 100m in reaching the opposite bank. The time taken now is 't'. If the same boat is to cross the river by shortest path, the time taken to cross will be
 1) 2t 2) $\sqrt{2}t$ 3) 3t 4) $\frac{2t}{\sqrt{3}}$
9. The speed of water in a river is 4 kmph and the speed of boat in still water is 8 kmph. If the same boat is used to travel a distance of 16km in the river from A to B downstream and back from B to A. Find the total time taken by the boat (in Hr)
 1) 8/3 2) 16/3 3) 4 4) 4/3
10. A man can swim with a certain speed, in still water to cross a river to a point directly opposite to the starting point in time t_0 . When the river flows, he crosses the river directly along the same path in a time t_1 . If 'd' be the river width, then the velocity of water current is
 1) $\frac{d\sqrt{t_1^2 - t_0^2}}{t_1 t_0}$ 2) $\frac{d\sqrt{t_1^2 + t_0^2}}{t_1 t_0}$ 3) $\frac{d(t_1 - t_0)}{t_1 t_0}$ 4) $\frac{d(t_1 + t_0)}{t_1 t_0}$
11. Wind is blowing in the north direction at speed of 2 m/s which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him:
 1) 2 m/s south 2) 2 m/s north 3) 4 m/s west 4) 4 m/s south

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. Three ships A, B & C are in motion. The motion of A as seen by B is with speed v towards north – east. The motion of B as seen by C is with speed v towards the north – west. Then as seen by A, C will be moving towards
 a) North b) South c) East d) West

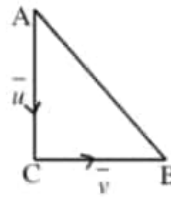
2. ABC is a right angled triangle, right angled at C. Two particles start from A and C, along AC and CB respectively with uniform velocities u and v . The shortest distance between them is ($AC=L$)

a) $\frac{Lv}{\sqrt{v^2 + u^2}}$

b) $\frac{Lu}{\sqrt{v^2 + u^2}}$

c) $\frac{Lv}{\sqrt{v^2 - u^2}}$

d) $\frac{Lu}{\sqrt{v^2 - u^2}}$



3. In the above problem the time after which the bodies are closest to each other is

a) $\frac{vL}{u^2 + v^2}$

b) $\frac{uL}{v^2 + u^2}$

c) $\frac{u}{\sqrt{v^2 + u^2}}$

d) $\frac{v}{\sqrt{v^2 + u^2}}$

4. A river flows at the rate of 3 km/hr and a person can row a boat at a speed of 5 km/hr in still water. If the difference between the times taken to cross the river by the shortest path and the quickest time be 4 minutes, find the width of the river.

a) $2\frac{1}{3}$ km

b) $1\frac{1}{3}$ km

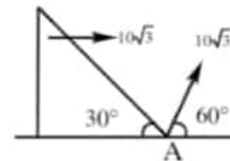
c) $3\frac{1}{4}$ km

d) $4\frac{1}{3}$ km

More than One correct answer Type Questions

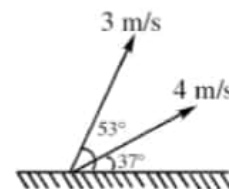
5. A particle projected at angle 60° with speed $10\sqrt{3}$, from the point 'A' as shown in the figure. At the same time the wedge is made to move with speed $10\sqrt{3}$ towards right as shown in the figure.

- a) time taken by particle to strike wedge is 2sec
b) Range on wedge is 20m
c) Maximum horizontal displacement $10\sqrt{3}$ m
d) Final velocity of particle is 10m/s



6. Two particles are projected with speed 4 m/s and 3 m/s simultaneously from same point as shown in the figure. Then:

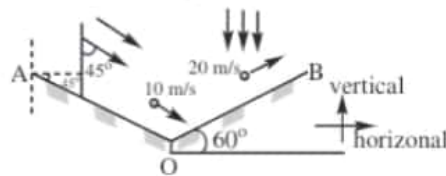
- a) Their relative velocity is along vertical direction
b) Their relative acceleration is non-zero and it is along vertical direction
c) They will hit the surface simultaneously
d) Their relative velocity is constant and has magnitude 1.4 m/s



7. A man running on a horizontal road at 8kmph finds that rain is falling vertically. He increases the speed to 12kmph and find that the drops make angle 30° with the vertical.

- a) The speed of rain is $8\sqrt{3}$ kmph
b) The speed of rain is $\sqrt{112}$ kmph
c) Speed of rain wrt man is 8kmph
d) Speed of rain wrt man is $8\sqrt{3}$ kmph

8. A swimmer crosses a flowing stream of width ' ω ' to and fro in time t_1 . The time taken to cover the same distance up & down the stream is t_2 . If t_3 is the time swimmer would take to swim a distance 2ω in still water, then:
- a) $t_1^2 = t_2 t_3$ b) $t_2^2 = t_1 t_3$ c) $t_3^2 = t_1 t_2$ d) $t_3 = t_1 + t_2$
9. A man swimming down stream overcome a float at a point M. After travelling distance D he turned back and passed the float at a distance of $D/2$ from the point M, then the ratio of speed of swimmer with respect to still water to the speed of the river will be
- a) 2 b) 3 c) 4 d) 2.5
10. A swimmer swims in still water at a speed = 5 km/hr. He enters a 200 m wide river, having river flow speed = 4 km/hr at point A and proceeds to swim at an angle of 127° with the river flow direction. Another point B is located directly across A on the other side. The swimmer lands on the other bank at point C, from which he walks the distance CB with a speed = 3 km/hr. The total time (In minutes) in which he reaches from A to B is
- a) 4 b) 1 c) 2 d) 6
11. To a man running upwards on the hill, the rain appears to fall vertically downwards with 4 m/s. The velocity vector of man with respect to the Earth is $(2\mathbf{i} + 3\mathbf{j})$ m/s. If the man starts running down the hill with the same speed, then the relative speed of the rain with respect to the man is
- a) $4\mathbf{i} + 2\mathbf{j}$ b) $-4\mathbf{i} - 2\mathbf{j}$ c) $4\mathbf{i} - 2\mathbf{j}$ d) $-4\mathbf{i} + 2\mathbf{j}$
12. There are two incline planes AO and OB inclined at 45° and 60° respectively with the horizontal as shown in the figure. When a man moves on the incline plane AO with speed 10 m/s, he observes that the rain drops are falling 45° with the vertical. When the man moves on the inclined plane OB with speed 20 m/s, he observes that the raindrops are falling vertically downward. Then actual speed of rain is



- a) $5\sqrt{2}$ m/s b) $20\sqrt{2}$ m/s c) $10\sqrt{2}$ m/s d) None of these

Linked Comprehension Type Questions

Passage-I :

An aeroplane takes flight from a place towards another place in a direction θ N of E in a strong wind of velocity U to the west. The speed of plane in still air is V . Distance between the two places is L .

13. If T_1, T_2 are time of travel of on ward and return journey, then $T_1 T_2 = \dots\dots\dots$

- a) $\frac{L^2}{v^2 + u^2}$ b) $\frac{L^2}{v^2 - u^2}$ c) $\frac{2L^2}{v^2 - u^2}$ d) $\frac{L^2}{2(v^2 - u^2)}$

14. At what angle the plane must be driven wrt line of flight

- a) $\sin^{-1}\left(\frac{u}{v}\sin\theta\right)$ b) $\sin^{-1}\left(\frac{v}{u}\sin\theta\right)$ c) $\cos^{-1}\left(\frac{u}{v}\sin\theta\right)$ d) $\tan^{-1}\left(\frac{u}{v}\sin\theta\right)$

Passage-II :

Consider the previous situation with $V_R > V_{bR}$. Let $V_R = V_1$; $V_{bR} = V_2$

15. For this given situation the person
- Can never reach the exactly opposite point A by swim.
 - May reach the exactly point, if $\theta \neq 0$
 - May (or) may not reach A depending on V_1 & V_2
 - Data insufficient to conclude whether person can reach opposite point A directly (or) not
16. For the drift (x-displacement) to be minimum, the angle θ at which the person should swim from O is
- $\sin^{-1}\left(\frac{V_{BR}}{V_R}\right)$
 - $\sin^{-1}\left(\frac{V_R}{V_{BR}}\right)$
 - $\cos^{-1}\left(\frac{V_{BR}}{V_R}\right)$
 - $\cos^{-1}\left(\frac{V_R}{V_{BR}}\right)$
17. The time taken by the swimmer to have minimum drift is
- $\frac{dV_1}{(\sqrt{V_1^2 - V_2^2})V_2}$
 - $\frac{d}{V_1}$
 - $\frac{d}{V_2}$
 - $\frac{d}{\sqrt{V_1^2 + V_2^2}}$

Matrix Matching Type Questions

18. Two particles A and B moving in x-y plane are at origin at $t = 0$. The initial velocity vectors of A and B are $U_A = 8\hat{i} \text{ m/s}$ and $U_B = 8\hat{j} \text{ m/s}$. Their accelerations are $a_A = -2\hat{i} \text{ m/s}^2$ and $a_B = -2\hat{j} \text{ m/s}^2$

Column-I

- The time in seconds at which relative velocity of A wrt B is zero
- The distance in metres between A and B when their relative velocity is zero
- The time in seconds at which they have same position
- The relative velocity in m/s when they are at same position

Column-II

- $16\sqrt{2}$
- $8\sqrt{2}$
- 8
- 4

19. A boat which has a speed of 5kmph in still water crosses a river of width 1km along shortest path in 15min. (river velocity is greater than 5kmph). In second coloumn velocities are in kmph and lengths are in Km, time is in hours

Column-I

- Velocity of river
- Shortest time taken to cross river
- Length of shortest path
- Minimum drift

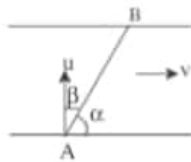
Column-II

- 4/3
- 5/3
- 25/3
- $\frac{1}{5}$

Integer Type Questions

20. A stone is dropped from a height h . Simultaneously another stone is thrown up from the ground with such a velocity that it can reach a height of $4h$. The time when two stones cross each other is $\sqrt{\left(\frac{h}{kg}\right)}$ where k is

21. Point A moves uniformly with velocity v so that velocity v is continually "aimed" at point B which in its turn moves rectilinearly and uniformly with velocity $u < v$. At the initial moment of time $v \perp u$ and the points are separated by a distance l . The time at which the two points will converge (if $v = 2u$) is $kl/3u$. The value of 'k' is
22. A launch travels between two points A and B on the opposite banks of a river always following the line AB. The distance S between points A and B is 280 m. The velocity of the river current $v = 6$ m/s is constant over the entire width of the river. The line AB makes an angle $\alpha = 53^\circ$ with the direction of the current. The angle β remains the same during the passage from A to B and from B to A. The velocity 'u' and angle β to the line AB should the launch to be driven to cover the distance AB and back in a time $t = 128$ sec is $2p$ m/sec and $37^\circ q$. The values of p and q are



23. A man can row a boat with 4 kmph in still water. If he is crossing a river where the current is 8 kmph. The time taken to cross the river in shortest path is $\frac{k}{\sqrt{3}}$ hr. where 'k' is (width of the river is 8 km)

EXERCISE-VIII

(Problems on Calculus)

LEVEL-I (MAIN)

Straight Objective Type Questions

- A particle starts from rest from origin and moves along a straight line according to the law $a = \mu \cos t$ where $a =$ acceleration and μ is a constant. Find its displacement in time t .
1) $S = \mu(1 + \cos t)$ 2) $S = \mu(1 - \cos t)$ 3) $S = \mu(1 + \sin t)$ 4) $S = \mu(1 - \sin t)$
- A particle moving in a straight line starts with a velocity u . Its retardation at any instant is proportional to the displacement. Find its displacement before coming to rest, if retardation at unit displacement is μ .
1) $S = \pm \frac{u}{\sqrt{\mu}}$ 2) $S = \pm \frac{2u}{\sqrt{\mu}}$ 3) $S = \pm \frac{u}{2\mu}$ 4) $S = \pm \frac{\mu}{u}$
- A train starting from rest is accelerated and the instantaneous acceleration is given by $\frac{10}{v+1} \text{ m/s}^2$. Where v is the velocity in m/s. Find the time after which the body attains a velocity of 54 km/hr.
1) 10.75 sec 2) 12.75 se 3) 12.25 sec 4) 10 sec
- A train starting from rest is accelerated and the instantaneous acceleration is given by $\frac{10}{v+1} \text{ m/s}^2$. Where v is the velocity in m/s. Find the distance in which the train attains a velocity of 54 km/hr.
1) 123.75 m 2) 112.5 m 3) 11.25 m 4) 77.25 m
- The relation between time t and distance x is $t = \alpha x^2 + \beta x$. Where α and β are constants. The retardation is
1) $2\alpha v^3$ 2) $2\beta v^3$ 3) $2\alpha\beta v^3$ 4) $2\beta^3 v^3$

EXERCISE-I

LEVEL - I

1) 2 2) 1 3) 1 4) 2

LEVEL - II

1) c 2) a 3) b 4) ad 5) acd 6) 1 7) 1 8) 2

EXERCISE-II

LEVEL - I

1) 4 2) 2 3) 2 4) 4 5) 3 6) 3 7) 4 8) 3
9) 4 10) 1

LEVEL - II

1) b 2) a 3) c 4) b 5) b 6) c
7) A-R;B-P;C-S;D-Q 8) 1 & 2

EXERCISE-III

LEVEL - I

1) 2 2) 4 3) 4 4) 2 5) 3 6) 3 7) 2 8) 4

LEVEL - II

1) c 2) a 3) b 4) abc 5) a 6) b 7) a
8) A-QS; B-QR; C-PR; D-PS

EXERCISE-IV

LEVEL - I

1) 3 2) 2 3) 2

LEVEL - II

1) b 2) a 3) 2 4) 2

EXERCISE-V

LEVEL - I

1) 3 2) 2 3) 3 4) 4 5) 1 6) 2 7) 3 8) 1
9) 1

LEVEL - II

1) c 2) b 3) d 4) a 5) d 6) a 7) a 8) d
9) c 10) bcd 11) bcd 12) abc 13) abcd 14) b 15) b 16) d
17) c 18) a 19) A-R;B-S;C-Q;D-P 20) A-R;B-S;C-Q;D-Q
21) 3

EXERCISE-VI

LEVEL - I

1) 1 2) 2 3) 2 4) 1 5) 4

LEVEL - II

1) b 2) a 3) b 4) c 5) a 6) d 7) c 8) 1

EXERCISE-VII

LEVEL - I

1) 1 2) 3 3) 2 4) 4 5) 1 6) 2 7) 1 8) 4
9) 2 10) 1 11) 2

LEVEL - II

1) b 2) a 3) a 4) b 5) a 6) b 7) a 8) a
9) c 10) abcd 11) cd 12) bc 13) b 14) a 15) a 16) a
17) a 18) A-S;B-P;C-R;D-Q 19) A-R;B-S;C-Q;D-P 20) 8
21) 2 22) 4 & 1 23) 4

EXERCISE-VIII

LEVEL - I

1) 2 2) 1 3) 2 4) 1 5) 1